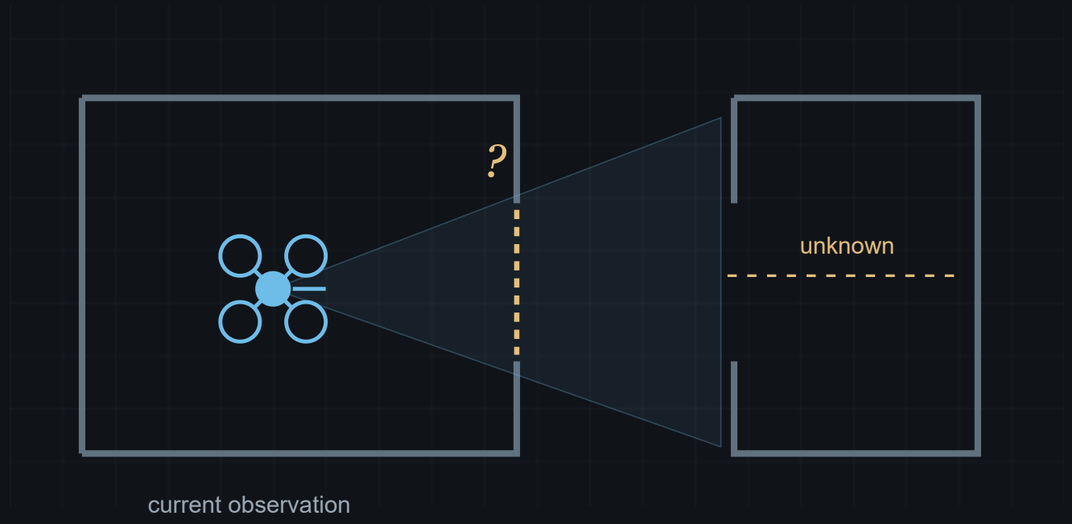
# The course is visible.
# The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.




*What should happen next?*

A passage candidate

What lies beyond it?

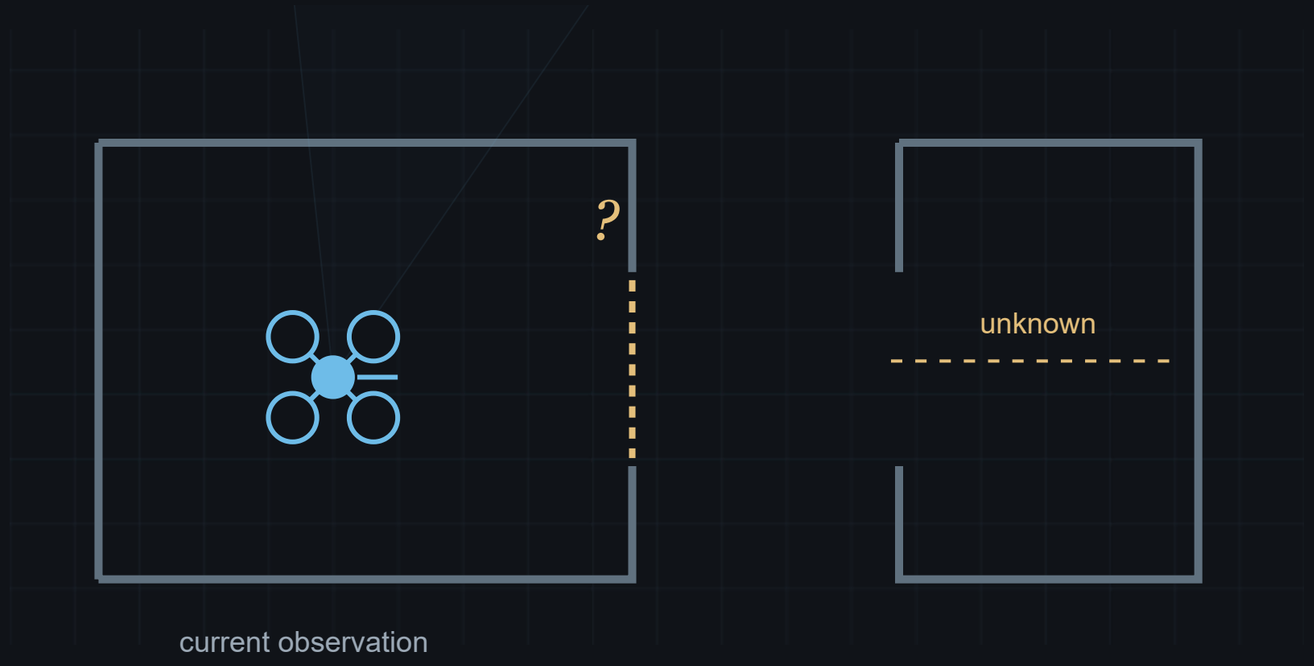
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

# Build what the drone *remembers*.

Persistent identity. Evidence. Uncertainty. A reason to observe again.




## *Memory ≠ a fresh view*

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.



Illustrative belief update

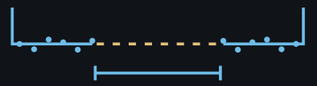
# Combine ideas at the interfaces.

Research gave us the structure; our constraints shaped the implementation.

RESEARCH → ADAPTATION

## Hydra

incremental spatial hierarchy

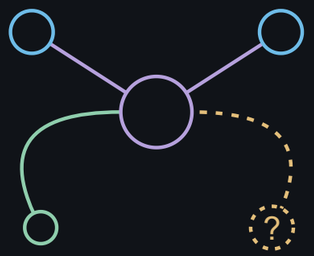


## Kimera

metric + semantic structure



## Compact WorldState



geometry + identity + evidence

CSIRO: recovery in partial observation

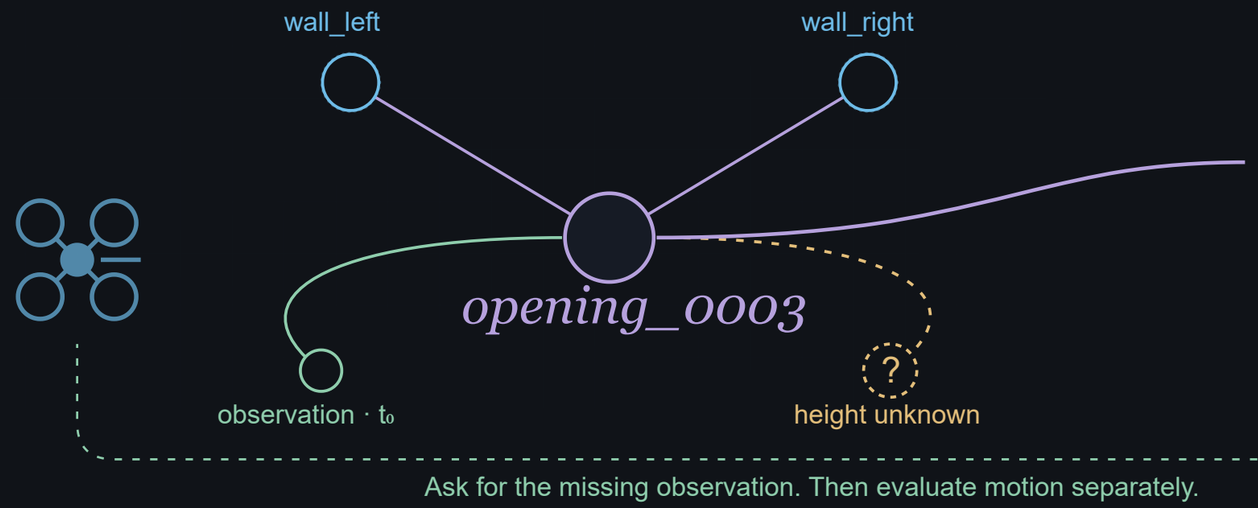
ModalAI: separate perception, planning and autopilot interfaces

↳ A compact adaptation for our task and compute budget.

# From seeing to *understanding*.

Observations become geometry, then meaning, then a grounded decision.

01 OBSERVATIONS | 02 GEOMETRY | 03 SEMANTIC GRAPH | 04 REASONING




GOAL PROPOSAL

## observe_more

target: opening_0003

because: height is unknown

Grounded proposal ≠ permission to fly

Ask for the missing observation. Then evaluate motion separately.

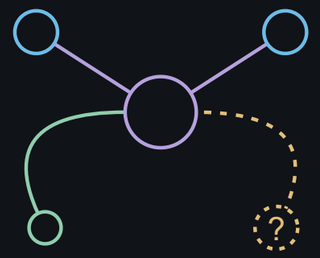
↳ Missing evidence becomes a reason to look again.



Illustrative entity · offline contract

# Sometimes the next action is *to look*.

A bounded reasoning layer turns missing information into an explicit goal.



*opening_0003*

height remains unknown

## Bound the context

entity + evidence

frame + time + missing facts

no invented connectivity

## Choose the next question

**observe_more**

target: opening_0003

because: height is unknown

## Grounding + freshness validated

Motion still needs its own feasibility check.

↳ “Observe this opening again” is a useful decision.

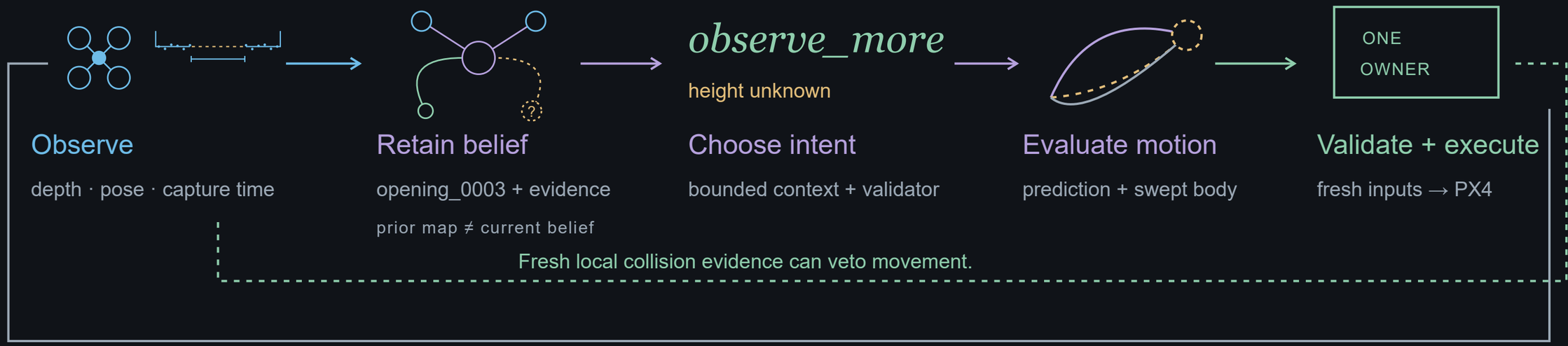


Non-executable proposal

# Each layer answers a different question.

Our target architecture separates memory, intent, motion and execution.

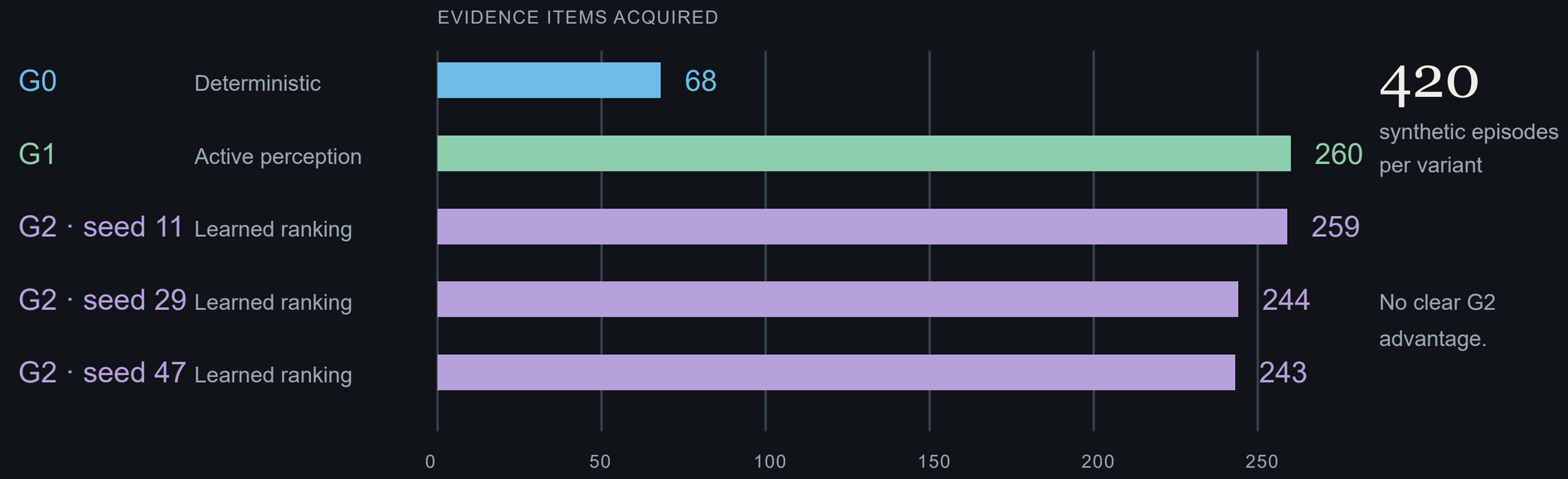
THE SAME EVIDENCE, THROUGH DIFFERENT RESPONSIBILITIES




↳ Persistent belief and fresh collision evidence have different jobs.

# Complexity has to earn its place.

Active perception earned priority in a synthetic evidence-acquisition task.




**420**
synthetic episodes per variant

No clear G2 advantage.

↳ Active perception improved this task; learned ranking showed no clear advantage.

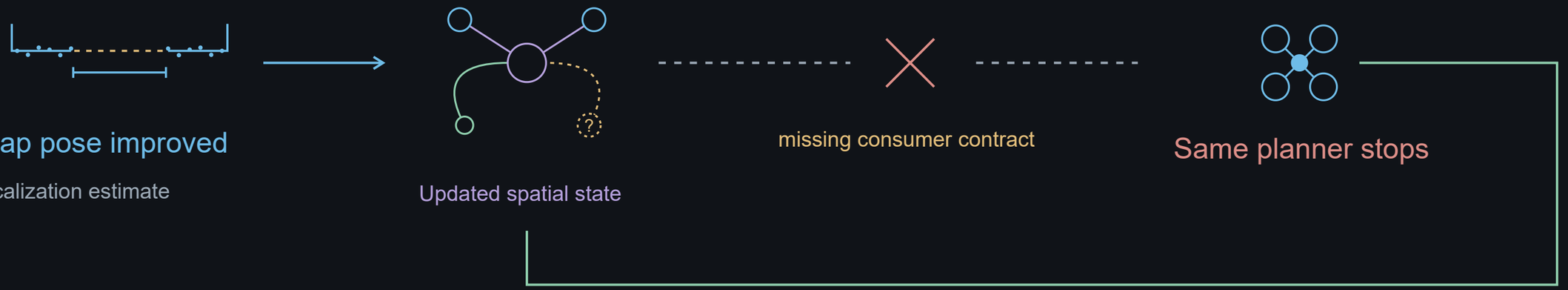


Synthetic benchmark · not flight success

# A better map was not yet a better decision.

An integration lesson: better estimates must reach the decisions they are meant to change.

HISTORICAL P2 FIXTURE

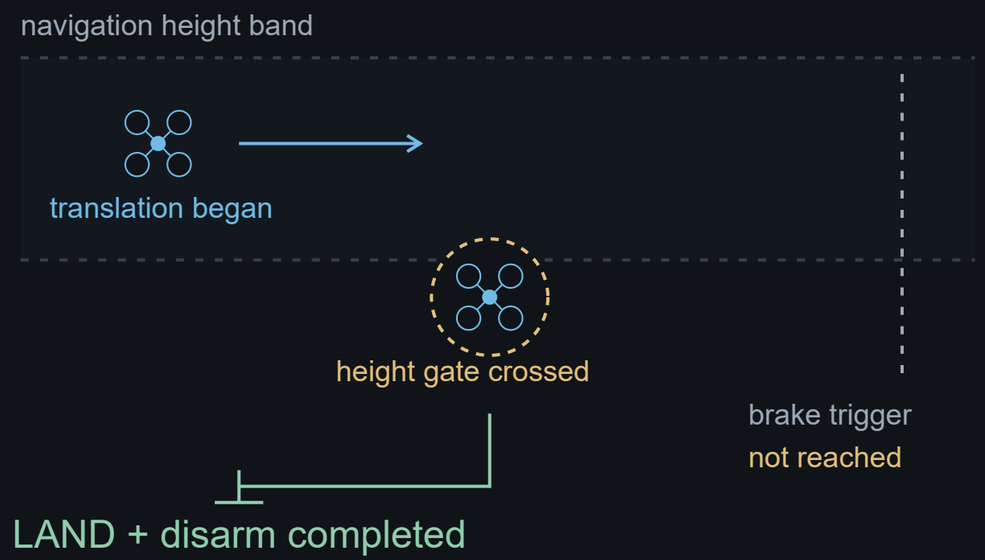



↳ We shifted toward explicit consumer contracts and bounded asynchronous registration.

# The assumptions failed between layers.

Native experiments exposed altitude, frame, timing and scheduling contracts.

C1 · THE BRAKING MEASUREMENT FAILED




Schematic states and event order · not a measured trajectory

S0 · FOUR DISARMED ATTEMPTS



**01 Frame origins**

bind the same physical position



**02 Clock alignment**

preserve capture-time identity



**03–04 Readiness + scheduling**

bound the work on the owner path

No course-flight success claimed.

↳ Preserved failures became specific engineering work.

# What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

EVIDENCE IS LOCAL TO THE CLAIM



## Offline components

entities · reasoning · mapping

synthetic comparison evidence



## Native calibration

C0: completed

C1: recovered; measurement failed




## Mission-level benefit

hybrid course + matched trials

hardware qualification remains

Separate baseline: Classical S1 has one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

# From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.



### 01 Reliable native segment

frames · timing · height control



### 02 Matched comparison

same map, start and scoring




### 03 Layer ablations

memory → topology → semantics

*Observations* ⟶ *Grounded belief* ⟶ *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Our contribution is a testable path from observations to justified actions.

N1 · N6 · N7 · N8

Planned evaluation

# A map can say where. Semantics asks what next.

## Three representations

- **Geometry:** surfaces, pose, observed free space and clearance.
- **Topology:** places and supported connections.
- **Semantics:** task-relevant meaning, such as a candidate passage.

## Why keep memory?

Retain identity when a feature leaves view. Track stale or contradictory evidence. Select an information-gathering goal rather than inventing a connection.

These are architectural capabilities; mission benefit still needs a matched ablation.

# Learning is a replaceable component.

## Reasoning comparison

- G0: deterministic baseline.
- G1: utility / information-gain selection.
- G2: compact learned candidate ranking, evaluated with three seeds.

## Motion prediction

D0 is the nominal physics default. Residual and adaptive alternatives exist offline, but clean-case improvements did not satisfy every delay, heading and runtime gate.
A learned predictor is not a learned navigation policy or a VLA controlling the aircraft.

# Compare the same task, under the same conditions.

## Reasoning benchmark

60 held-out layouts × 7 regimes × 6 steps. 420 episodes per variant.
Evidence acquired: G0 68; G1 260; G2 259 / 244 / 243.
Abstract synthetic evidence acquisition, not course completion.

## Separate Classical S1 result

One Gazebo run: four checkpoints, pad landing and disarm in 84.3 seconds; zero touches reported.
Run `run-AdMalPxA`, revision `ad31ae5`. Not a matched hybrid comparison or physical-hardware result.

# A useful fact needs a frame and a clock.

## Data contracts

- Capture time, frame, units and reset epoch.
- Stable identity and supporting observations.
- Bounded context and independent validation.
- One command owner.

## Four separate layers

- Permitted reference geometry.
- Persistent current-run belief.
- Fresh local collision evidence.
- Semantic and topological state.

A prior map does not become a fresh sensor observation.

# A recovered experiment can still be a failed test.

## Preserved outcomes

C1 crossed its height threshold before the brake trigger. LAND and disarm completed, but no braking envelope was measured. Latest fetched S0 attempts 001–004 stayed disarmed. Frame, timestamp and scheduling work remained.

## Acceptance sequence

Native segment → matched baseline → memory / topology / semantics ablations → changed-layout evaluation.
Measure mission progress, false-feasible choices, observation cost and latency. Add learned complexity when these improve.

# The ideas we adapted.

## Academic references

Hydra · Hughes et al. · RSS 2022
Incremental hierarchical scene graphs.

Kimera · Rosinol et al. · 2021
Metric-semantic spatial representation.

CSIRO Data61 · Hudson et al. · 2021
Fielded subterranean robotic systems.

## Engineering and visual references

ModalAI VOXL Vision Hub
VIO, obstacle processing and autopilot interfaces.

3Blue1Brown / Manim
Explanatory animation inspiration. All diagrams here are original SVG illustrations.

Methods inform our design; their published results do not validate our platform.